

WORK EXPERIENCE

American Express

July 2023 – July 2024

Associate – Digital Product Management (Global Risk & Compliance Dept)

- **Saved ~\$70K** in Q1–Q2'24 through 33% data rationalization, by designing & developing **Python** and **SQL** pipelines to identify and resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team.
- Led the design and implementation of onboarding pipelines for **American Express's Kabbage acquisition**, migrating **~20K small-business accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance and data reliability.
- Maintained, debugged production systems, & optimized production data layers, by tracking telemetry and data integrity metrics through **Yellowbricks** and collaborating cross-functionally in an Agile environment via **JIRA**, communicating status and resolving issues with the AML Transaction Monitoring team.
- Streamlined onboarding for new AML data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

American Express

May 2022 – July 2022

Analyst Intern - Data Engineering (Credit & Fraud Risk Dept)

- Reduced development time by 80% for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios.
- Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.

TECHNICAL SKILLS

Language: C, C++, Python, MATLAB, PHP R, SQL, PostgreSQL, HiveQL, TypeScript, JavaScript**Cloud Tech:** Amazon Web Services: EC2, S3, Lambda, ECS, CodePipeline; Google Cloud : GCP, Vertex AI, Firestore.**DevOps, Infra & Delivery:** Git, Docker, Kubernetes, Apache Kafka, Redis, Linux OS, Slurm, SDLC, CI/CD, Jenkins.**App Dev & Analytics:** HTML, CSS, REST APIs, React, Node.js, FastAPI, Uvicorn, Streamlit, n8n, PHPUnit.**ML & AI:** scikit-learn, numpy, scipy, pandas, LangChain, keras, tensorflow, LangGraph, transformers; BERT, Gemini 2, RoBERTa, all-MiniLM-L6-v2, all-mpnet-base-v2, VGG19, Opacus, Presidio, GPT-2, NetworkX.**AI Coding Tools:** Claude Code, Codex, Cursor, Github Copilot**Coursework:** Data Structures, Algorithms, Object Oriented Programming, Software Engineering, System Design, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Blockchain, Natural Language Processing.

EDUCATION

University of Michigan, Ann Arbor

Aug. 2024 – May 2026

*Master of Science in Engineering in Computer Science***GPA: 4.0/4.0****Delhi Technological University, Delhi**

July 2019 – May 2023

*Bachelor of Technology in Mathematics & Computer Science***GPA: 3.81/4.0**

PUBLICATION

“University Recommendation System” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)** on 6–7th May 2023 at Gurgaon, India.

PROJECT EXPERIENCE

Wikispeech (MediaWiki extension): Open Source Contribution

- Built Wikimedia's article-to-audio download feature, enabling users to listen to and **save Wikipedia articles as speech**; delivered 4 **PHP** patches spanning schema design, **REST APIs**, JobQueue processing, and UI workflows with 40+ **PHPUnit** tests.

CineRank: Real-Time Movie Recommendation System

- Built a real-time recommendation engine that continuously updates user preferences & serves movie suggestions in **<100ms latency** on the **MovieLens 32M** dataset. Designed a streaming ML pipeline using **Kafka**, **Redis**, **PostgreSQL**, **FastAPI**, combining SVD collab filtering, cold-start fallback logic, & cached recommendation retrieval.

Needle: RAG-based Research Assistant

- Built an RA for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & uploaded PDFs, removing hallucinated citations through retrieval-backed generation. Developed a RAG pipeline using **Streamlit**, **Vertex AI**, **Gemini 2**, & **LangChain** with semantic search, citation verification, & conversational document querying.

PrivNLGBench: Differential Privacy Benchmark

- Built a benchmark for evaluating privacy-preserving NLP systems by transforming real-world datasets (Yelp, Enron, Clinton FOIA) into structured test sets with isolated PII and PII-free summaries. Developed the evaluation pipeline using **PyTorch**, **Opacus**, **DistilRoBERTa**, & **NetworkX**, achieving **0.85 accuracy** / **0.71 macro-F1** for PII detection and improving calibration from **0.272 to 0.083 ECE**.

MiniGPT: Built a decoder-only GPT-style **transformer** from scratch using **pytorch** & **numpy** with multi-head self-attention & positional embeddings, pretrained on web text & fine-tuned for multi-turn dialogue & MCQ answering.

AI Generated Text Detector: Built a logistic regression classifier using **scikit-learn** & **scipy** sparse features, with ~80% accuracy on AI-written text detection, benchmarking SGD, RMSprop, & AdamW optimizers in **pytorch**.

Chaos Classification on Keyword Extraction: Built a novel chaos classifier using Henon maps, Lyapunov exponents on **scikit-learn** tf-idf, **spacy** NER, **NLTK** tokenization, & word2vec (**GloVE**) features to extract chaotic keywords from a corpus.

COVID-19 Detection using CNN: Built a **python** lung X-ray classification system using **tensorflow** and **keras** with VGG19 and ResNet50 architectures, achieving 95% accuracy on COVID-19 detection.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Generation Women Leaders 2023 & 2025 | **URAM (Micron) Scholarship** 2022 | **AWS ML Scholarship** 2021 | UCMAIS International Distinction Grade 10
- **Leadership:** Judge, MHacks 17 (MLH Official) | **Vice Chairperson**, IEEE DTU CS | Student Advisory Board, Michigan Engineering | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & American Express, Joy of Coding (UMich)